

SMARTBRIDGE CAPSTONE

LEGALEASE: AI-Powered Document Generator

Automating tailored, legal-grade contracts, NDAs, and agreements using Gemini 1.5 Pro & FastAPI.

CORE TECH

FastAPI + Streamlit + Gemini 1.5

OUTPUT FORMATS

Branded PDF, DOCX & TXT

Project Purpose & Core Vision

Bridging Legal Complexity

Connects complex legal requirements with effortless operational accessibility.

Instant Contract Generation

Empowers non-lawyers to create sound contracts without traditional friction.

Custom Branding & Standards

Embeds logos, custom fonts, clause tables, and standard export formats.



Real-World Deployment Scenarios



Startup Founder

Employment Contracts

Automates multi-clause hiring agreements and IP assignment.

Output: Branded PDF



Freelancer

Non-Disclosure (NDA)

Generates confidentiality agreements with custom terms.

Output: Fast TXT / DOCX



Landlord

Residential Leases

Drafts property lease agreements with structured tables.

Output: Word (.DOCX)

Technology Stack & Frameworks

FastAPI Framework

High-performance Python backend serving REST routes and Pydantic validation.

Google Gemini 1.5 Pro

Generative AI core with 1M token context for long-form legal reasoning.

Streamlit Frontend

Responsive interface for parameters, live HTML preview, and inline editing.

FPDF & Python-Docx

Document formatting engines embedding logos, headers, and structured tables.

System Architecture & Data Pipeline

01

Streamlit UI

Captures parties,
document type, terms,
and dates.



02

FastAPI Engine

Validates payload and
invokes generation code.



03

Gemini 1.5 Engine

Generates structured
clauses with formal
legalese.



04

Exporter Engine

Converts output to styled
PDF, DOCX, and TXT.

AI Integration: Gemini 1.5 Pro

1M

Token Context Window
Handles lengthy legal inputs

100%

Structured Prompting
Enforces mandatory sections

< 3s

Generation Time
Instantaneous output

Legal Domain Adaptability: Splits user inputs into structured bullet points and clause tables automatically.

Development Roadmap & Milestones

Milestone 1

Model Selection

Evaluated Gemini 1.5 Pro architecture.

Milestone 2

Core Features

Built FastAPI endpoints & handlers.

Milestone 3

API Logic

Integrated Pydantic schemas & API routes.

Milestone 4

Frontend UI

Created Streamlit app with live edit.

Milestone 5

Deployment

Tested local server launch & file exports.

Interactive UI & Export Capabilities

- 1 Parameter Input**
Input document category, parties, terms, and effective dates.
- 2 Real-Time Inline Editing**
Review generated clauses in a preview box with live edits.
- 3 Multi-Format Downloads**
One-click exports for .TXT, Word .DOCX, and styled PDF files.



Strategic Value & Industry Impact

10×

Speed Improvement

Reduces document creation time from hours to seconds.

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Cost Reduction

Minimizes expensive legal consultations for standard documents.

100%

Standard Consistency

Ensures mandatory legal clauses and uniform formatting.

EMPOWERING LEGAL ACCESSIBILITY

LegalEase bridges technology and law to make reliable document generation accessible.

Transparent, automated, and professional legal workflows for everyone.

Thank You

Q & A Session

Image Sources



https://elements-resized.envatousercontent.com/elements-video-cover-images/814ad124-d999-47ab-97b4-3a9be1155d4a/video_preview/video_preview_0000.jpg?w=500&cf_fit=cover&q=85&format=auto&s=ec6454b3b40d51d55a5d41a3125c3b4a8d4f8a7a751f241d5ca7308f989ffe9d

Source: elements.envato.com



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Source: unsplash.com